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BALLARI INSTITUTE OF TECHNOLOGY & MANAGEMENT

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(Recognized by Govt. of Karnataka, approved by AICTE, New Delhi & Affiliated to Visvesvaraya Technological University, Belagavi)
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DEPARTMENT OF CSE(DATA SCIENCE)

A Mini Project Report On

"STUDENT PERFORMANCE DATA VISUALIZATION DASHBOARD"

A report submitted in partial fulfillment of the requirements for the

DATA VISUALIZATION WITH POWER BI

Submitted By

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Belagavi, Karnataka 2025-2026

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CERTIFICATE

This is to certify that the open ended project report on DATA VISUALIZATION WITH POWER BI **"STUDENT PERFORMANCE DATA VISUALIZATION DASHBORD"** has been successfully presented by **A ROHAN SUDHAN** bearing USN **3BR23CD002** student of semester B.E for the partial fulfillment of the requirements for the award of **Bachelor Degree in CSE(DS)** of the BALLARI INSTITUTE OF TECHNOLOGY& MANAGEMENT, BALLARI during the academic year 2025-2026.

Signature of guide	Signature of HOD
Mr. Sundeep Sindhanoor	**Dr. Aradhana D**

ACKNOWLEDGEMENT

The satisfactions that accompany the successful completion of open ended mini project on **"STUDENT PERFORMANCE DATA VISUALIZATION DASHBOARD"** would be incomplete without the mention of people who made it possible, whose noble gesture, affection, guidance, encouragement and support crowned my efforts with success. It is our privilege to express our gratitude and respect to all those who inspired us in the completion of our mini-project. We are extremely grateful to our Guides **Mr.Sundeep Sindhanoor** for their noble gesture, support co-ordination and valuable suggestions given in completing the mini-project. We also thank **Dr. Aradhana D,** H.O.D. Department of CSE(DS), for his co-ordination and valuable suggestions given in completing the mini-project. We also thank the Principal, Management and non-teaching staff for their co-ordination and valuable suggestions given to us in completing the project.

Name	USN
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ABSTRACT

This project focuses on developing an interactive student performance dashboard using Power BI to analyze academic and lifestyle data. The dataset consists of multiple interconnected tables, including Students, Academics, Exams, Lifestyle, and a derived DeptSummary table. The objective is to transform raw data into meaningful insights that help evaluate academic performance, lifestyle impact, and departmental trends. By integrating data using StudentID as a primary key, the system enables comprehensive analysis across multiple dimensions.

The dashboard presents key metrics such as average CGPA, study hours, sleep duration, screen time, and stress levels through intuitive visualizations. Correlation analysis is used to examine relationships between lifestyle factors and academic performance, while interactive charts and filters allow dynamic exploration of data. The findings indicate that student performance is influenced by multiple factors rather than a single variable, highlighting the importance of a balanced approach to academic and lifestyle activities.

Student Performance Data Visualization Dashboard Report

Chapter 1: Introduction

1.1 Background

Educational institutions generate large volumes of data, including academic scores, attendance, study habits, and lifestyle patterns. Analyzing this manually is inefficient. Power BI enables transforming this data into interactive dashboards for better insights.

1.2 Problem Statement

Traditional systems store student data in isolated tables and lack analytical capabilities. This makes it difficult to identify performance patterns or understand the influence of lifestyle factors on academic outcomes.

1.3 Objectives

- Establish relationships between datasets using StudentID
- Analyze academic performance across departments
- Study the relationship between lifestyle factors and CGPA
- Enable dynamic filtering using slicers

1.4 Scope

The system is designed for:

- Students
- Faculty
- Academic administrators
- Data analysts

Chapter 2: Dataset Description

2.1 Students Table

Contains basic student details such as:

- StudentID
- Name

- Department
- Year

2.2 Academics Table

Stores subject-level performance:

- StudentID
- Subject
- Marks
- Attendance

2.3 Exams Table

Contains academic results:

- StudentID
- CGPA
- Internal Marks
- External Marks

2.4 Lifestyle Table

Captures behavioral data:

- StudentID
- Study Hours
- Sleep Hours
- Screen Time
- Stress Level

2.5 DeptSummary Table

Derived table containing aggregated metrics:

- Avg CGPA
- Avg Study Hours
- Avg Sleep
- Avg Screen Time
- Correlation values

Chapter 3: Tools and Technologies Used

- **Power BI** → Dashboard creation
- **CSV/Excel** → Data storage
- **DAX (Data Analysis Expressions)** → Measures and calculations
- **Data Modeling** → Relationship building

Chapter 4: Data Modeling

The following relational flow was established to enable cross-table analysis:

Students → Academics → Exams → Lifestyle

- **StudentID acts as the primary key**
- **Enables cross-table analysis**

The model integrates academic and lifestyle data to enable multidimensional analysis of student performance.

Chapter 5: Data Preprocessing

Key steps performed:

- Imported datasets into Power BI
- Removed inconsistencies and duplicates
- Standardized data types
- Established relationships using StudentID
- Created calculated columns and measures

Correlation Measures:

- Corr Study vs CGPA
- Corr Sleep vs CGPA
- Corr Screen vs CGPA
- Corr Stress vs CGPA

Purpose:

To quantify relationships between lifestyle factors and academic performance.

Chapter 6: DAX Measures Created

Key Measures:

- Avg CGPA → AVERAGE(exams[CGPA])
- Avg Study Hours → AVERAGE(lifestyle[StudyHours])
- Avg Sleep → AVERAGE(lifestyle[SleepHours])
- Avg Screen Time → AVERAGE(lifestyle[ScreenTime])

Chapter 7: Dashboard Visualizations

- **KPI Cards**:
  - Avg CGPA → **7.95**
  - Avg Sleep → **6.75 hrs**
  - Avg Study Hours → **4.88 hrs**
  - Avg Screen Time → **4.38 hrs**
  - Avg Stress Level → **6.25**
- **Pie Chart**:
  - Distribution of students across departments
- **Bar Chart**:
  - Average CGPA comparison across departments
- **Scatter Plots**:
  - Study Hours vs CGPA
  - Sleep vs CGPA
  - Screen Time vs CGPA
  - Stress vs CGPA
- **Slicer**:
  - Department-wise filtering

Chapter 8: Dashboard Insights

• Average CGPA is 7.95, indicating stable academic performance across students.
• Students study 4.88 hours per day and spend 4.38 hours on screen time, showing a balanced lifestyle.
• Average sleep is 6.75 hours, slightly below recommended levels, while stress is moderately high at 6.25.
• CGPA varies across departments, with CS-AI and ECE performing better than others.
• Correlation between lifestyle factors and CGPA is very weak, indicating no strong direct impact.
• Students maintain consistent performance despite moderate study hours and stress levels.

Chapter 9: Benefits of the System

- Enables data-driven academic analysis
- Helps identify performance trends across departments
- Supports better understanding of student behavior
- Improves decision-making using visual insights

Chapter 10: Limitations

- Dataset is static (no real-time updates)
- Limited number of variables
- Weak correlation reduces predictive capability
- No advanced machine learning models

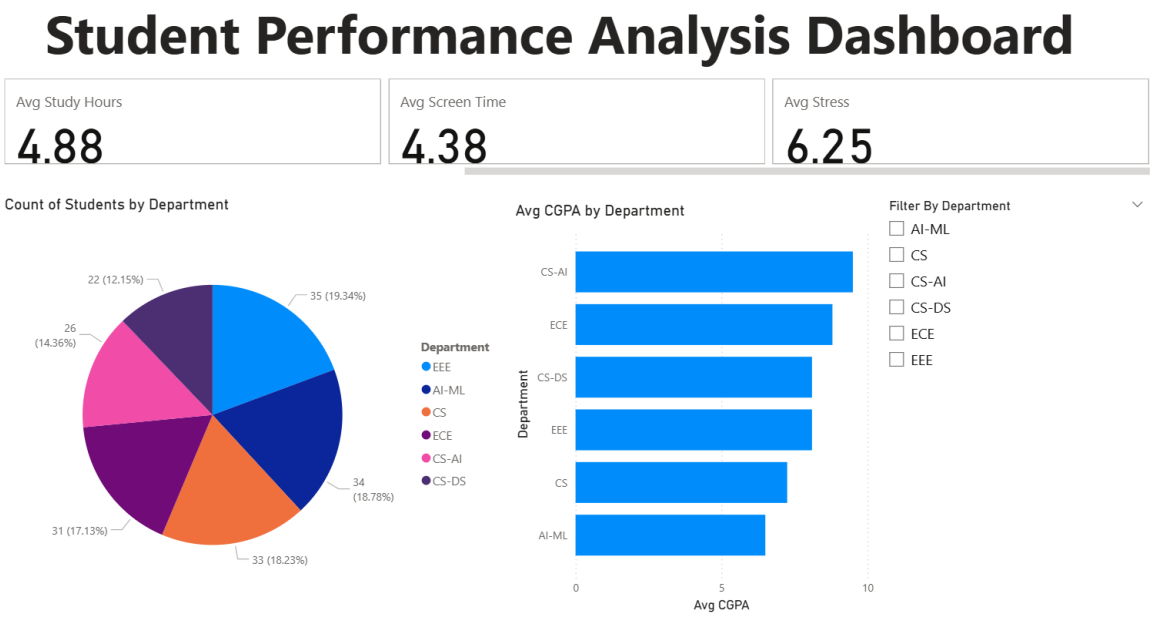
Chapter 11: Results

This dashboard provides a centralized analytical platform to study student performance.

Key Results:

- Academic performance is fairly consistent across departments

- No strong direct relationship between lifestyle variables and CGPA
- Data suggests complex multi-factor influence on performance
- Visualization helps simplify complex data interpretation



Key Insights from the Dashboard

Overall Student Behavior

- Students study an average of **4.88 hours per day**, indicating a moderate level of academic effort.
- Average screen time is **4.38 hours per day**, which is close to study time, suggesting a **balanced academic and digital lifestyle**.
- The average stress level is **6.25**, indicating **moderate to high stress among students**.

Department Distribution

- Student distribution across departments is **fairly balanced**, with no single department dominating the dataset.
- EEE has the highest representation (~19.34%), while CS-DS has the lowest (~12.15%).

Insight:

The dataset is well distributed, making comparisons across departments reliable.

Academic Performance by Department

- **CS-AI** has the highest average CGPA, indicating strong academic performance.
- **ECE** follows closely, showing consistent results.
- **AI-ML** has the lowest average CGPA among departments.

Insight:

Academic performance varies across departments, suggesting differences in curriculum difficulty, student behavior, or evaluation patterns.

Lifestyle vs Performance Observation

- Despite **moderate study hours**, students are able to maintain stable performance levels.
- Screen time being close to study hours suggests that **digital usage may not be entirely detrimental**.

Insight:

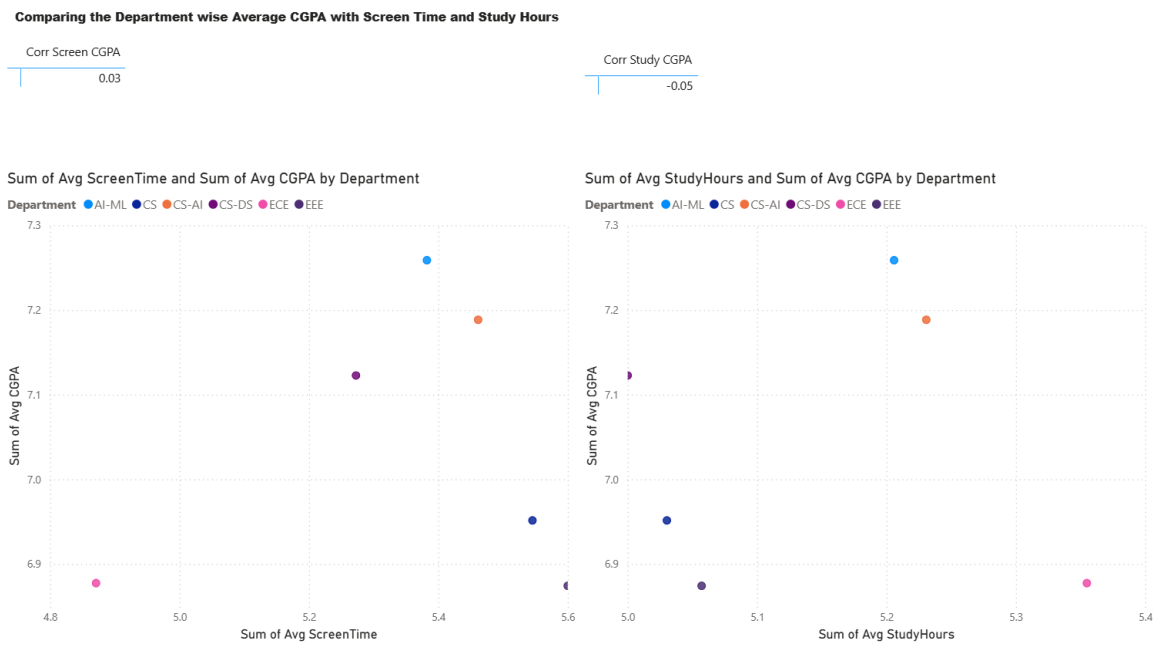
Students appear to maintain a balance between academic and non-academic activities without significantly impacting performance.

Stress Insight

- With an average stress level of **6.25**, students are experiencing noticeable academic pressure.

Insight:

Even with moderate study hours, students report relatively high stress, indicating possible external pressures or workload intensity.



Insights

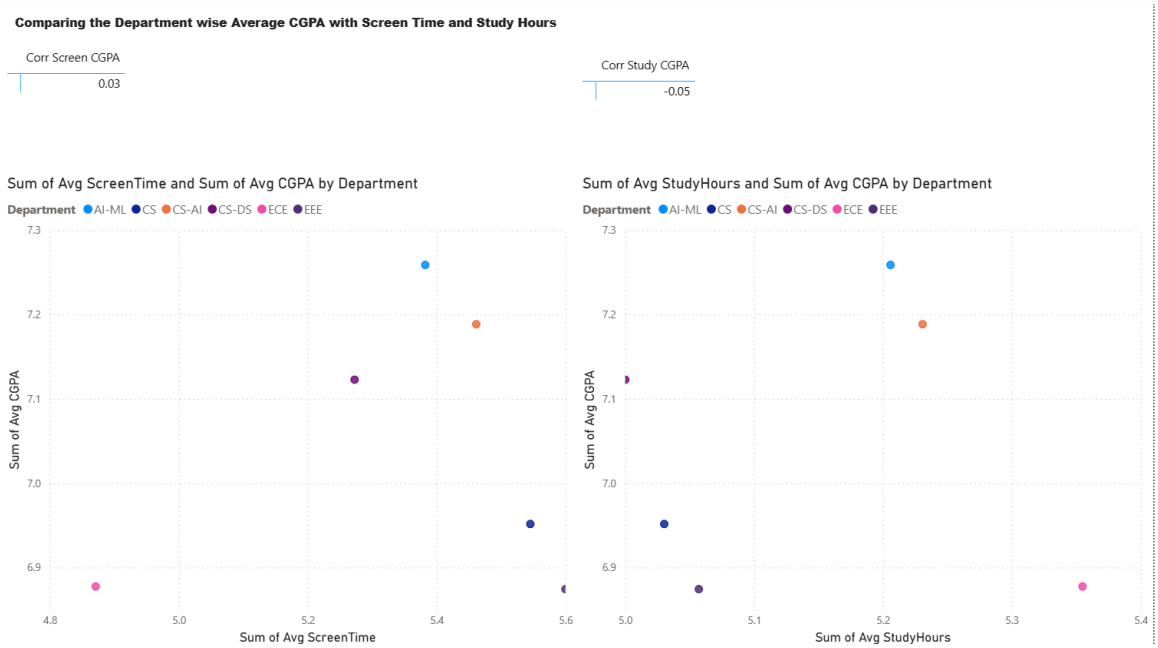
- Correlation between **screen time and CGPA (0.03)** and **study hours and CGPA (-0.05)** is very weak, indicating no strong relationship.
- Increasing study hours does not consistently improve CGPA, suggesting that **study efficiency matters more than duration**.
- Screen time does not show a clear negative impact on performance, indicating it may include **both productive and non-productive usage**.
- CGPA varies across departments despite similar study and screen time levels, showing that **performance depends on multiple factors**.

Department-Level Observations

- One department (AI-ML) shows relatively higher CGPA with moderate screen time and study hours.
- Some departments with higher screen time or study hours still have lower CGPA.

Insight:

Academic performance varies across departments and is likely influenced by multiple factors such as curriculum, learning approach, or evaluation methods.



Insights

- Correlation values are very low (**Screen: 0.03, Study: -0.05**), indicating **no strong relationship with CGPA**.
- Scatter plots show that departments with higher study hours or screen time do not consistently achieve higher CGPA.
- Most departments cluster within a narrow CGPA range (~6.8–7.3), showing **similar performance levels**.

Study hours and screen time alone do not significantly impact academic performance; results are influenced by multiple factors.

Chapter 12: Conclusion

The Student Performance Dashboard demonstrates how data visualization can enhance educational analytics. By integrating academic and lifestyle datasets, the system provides meaningful insights into student performance patterns and highlights the importance of data-driven decision-making.

Chapter 13: Future Enhancements

- Integration with real-time student data systems

- Predictive modeling for CGPA forecasting
- Machine learning for performance prediction
- Mobile dashboard deployment